

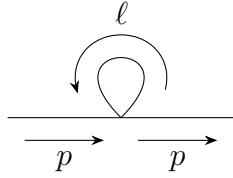
1. For $g = 0$ we have the following Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

In $d = 3$, the kinetic term gives us the dimension of the field $[\phi] = \frac{1}{2}$, therefore $[m] = 1$ and $\lambda = 1$, which means this theory is *super-renormalizable*. Using Weinberg's power counting rule, we have

$$[\Gamma_n] = 3 - \frac{1}{2}(\#\text{external lines}) - (\#\text{vertices})$$

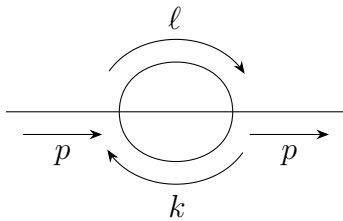
Given the interaction $\propto \phi^4$, the irreducible graphs can only have an even number of external lines. Therefore, for Γ_4 the only divergent irreducible graph is with one vertex; however, this is a three-level graph, and therefore it's convergent. The only graphs which are divergent are therefore the 2-point functions with one and two vertices:



1-loop contribution

$$[\Gamma_2^{(1)}] = 3 - \frac{1}{2}2 - 1 = 1$$

$$\Gamma_2^{(1)} = C^{(1)} + \Gamma_2^{(1)\text{reg}}$$



2-loop contribution

$$[\Gamma_2^{(2)}] = 3 - \frac{1}{2}2 - 2 = 0$$

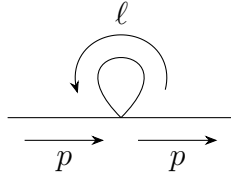
$$\Gamma_2^{(2)} = C^{(2)} + \Gamma_2^{(2)\text{reg}}$$

Overall, only the mass term Z_m has to be modified and only in the first and second order in perturbation λ , which corresponds to this theory being *super-renormalizable*.

In $d = 4$, we have $[\phi] = 1$, and therefore $m = 1$ and $\lambda = 0$, which means this theory should be *renormalizable*. Again, using Weinberg's power counting rule, we have

$$[\Gamma_n] = 4 - (\#\text{external lines})$$

From this relation, we see that all 1PI graphs with less than or equal to 4 are divergent while graphs with more than 4 external lines are convergent. Specifically, only the non-tree-level graphs with 4 external legs are divergent. In this case, the 2-point functions have the superficial degree of divergence $[\Gamma_2^{(l)}] = 2$, which causes both the mass term Z_m and the kinetic term Z_ϕ to have to be modified. These modifications start in the first order of perturbation:

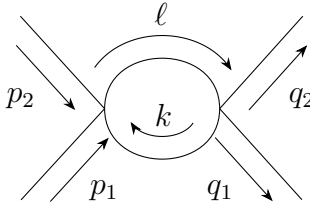


1-loop contribution

$$[\Gamma_2^{(1)}] = 4 - 2 = 2$$

$$\Gamma_2^{(1)} = C_m^{(1)} + C_\phi^{(1)} p^2 + \Gamma_2^{(1)\text{reg}}$$

Similarly, for all 1PI graphs with two external legs and more vertices, resulting in an infinite perturbation series expansion in Z_m and Z_ϕ . However, in the case of $d = 4$, the non-tree-level 1PI graphs with four external legs are also divergent, starting at one-loop:



1-loop contribution

$$[\Gamma_4^{(1)}] = 4 - 4 = 0$$

$$\Gamma_4^{(1)} = C_\lambda^{(1)} + \Gamma_4^{(1)\text{reg}}$$

In this case, the Z_λ term in the Lagrangian must be modified, again in an infinite order in perturbation.

[2.] In the case of a general g, λ in $d = 6$, we have the following Lagrangian density

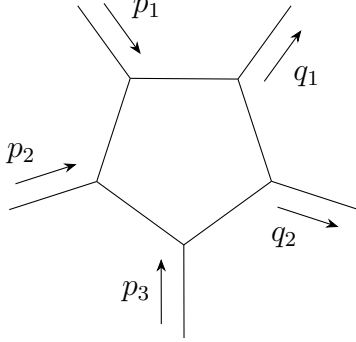
$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3 - \frac{\lambda}{4!}\phi^4.$$

In $d = 6$, the kinetic term gives us $[\phi] = 2$, which results in $[m] = 1$, $[g] = 0$ and $[\lambda] = -2$, which means that this theory for $\lambda \neq 0$ should be *non-renormalizable*.

In this generic theory, 1PI graphs with both an even and odd number of external legs are allowed. In general, we have

$$[\Gamma_n] = 6 - 2(\#\text{external lines}) + 2(\#\lambda\text{-type vertices})$$

From this expression, we see that the only convergent 1PI non-tree-level graphs have to have $(3 + n_\lambda + 1)$ external legs, where n_λ is the number of λ -type vertices in the graph, for example:



$$[\Gamma_5^{(1)}] = 6 - 2 \cdot 5 + 2 \cdot 0 = -4$$

$$\Gamma_5^{(1)} = \Gamma_5^{(1)\text{reg}}$$

However, for each n -point function, we can find a 1PI graph with a high enough number of λ -vertices such that the n -point function is divergent. Therefore, all operators, even the ones now written with zero coefficients in the Lagrangian density, will need modification in infinite order in perturbation.

[3.] In $d = 3$ the kinetic term gives us $[\phi] = \frac{1}{2}$, $[m] = 1$. For the Lagrangian to respect the global $\mathbb{Z}_2$ symmetry, we can only have terms which are even in the powers of the field ϕ . In addition, for the theory to be renormalizable, the operators in the Lagrangian can have at most order $[\mathcal{O}] = d = 3$. This results in only two interaction operators ϕ^4 and ϕ^6 . In terms of power counting and $\mathbb{Z}_2$ symmetry, the term $(\partial\phi)\phi^3$ could also come to mind; however, these terms could be written as a total four-divergence $\frac{1}{4}\partial(\phi^4)$ and therefore would not change the action. Overall, we obtain the following Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 - \frac{\kappa}{6!}\phi^6 - \Omega_0.$$

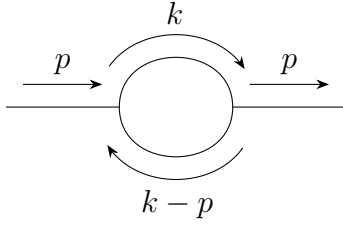
[4.] For $\lambda = 0$ the Lagrangian has the following form

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3.$$

In $d = 6$ we have $[\phi] = 2$, $[m] = 1$ and $[g] = 0$, and therefore this theory should be *renormalizable*. Specifically, the Weinberg's power-counting gives us

$$[\Gamma_n] = 6 - 2(\#\text{external lines}),$$

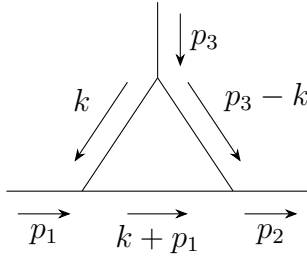
which results in Γ_2 being quadratically divergent and Γ_3 being logarithmically divergent. Specifically, for one-loop 1PI graphs, we have:



1-loop contribution

$$[\Gamma_2^{(1)}] = 6 - 2 \cdot 2 = 2$$

$$\Gamma_2^{(1)} = C_m^{(1)} + C_\phi^{(1)} p^2 + \Gamma_2^{(1)\text{reg}}$$



1-loop contribution

$$[\Gamma_3^{(1)}] = 6 - 2 \cdot 3 = 0$$

$$\Gamma_3^{(1)} = C_g^{(1)} + \Gamma_3^{(1)\text{reg}}$$

Overall, we see that for the 2-point function, the coefficients Z_m and Z_ϕ will have to be modified, and for the 3-point function, we will have to modify the Z_g coefficient.

Firstly, let us isolate the divergent term in the 2-point function. Let us denote

$$i\Gamma_2^{(1)}(p) = i(p^2 - m^2 + \Sigma(p))$$

and now we compute

$$\begin{aligned} i\Sigma_2(p) &= \frac{1}{2}(-ig)^2 \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i0} \frac{i}{(p-k)^2 + m^2 + i0} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 - m^2 + i0} \frac{1}{(p-k)^2 + m^2 + i0} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k}{(2\pi)^d} \frac{1}{[k^2 + x(1-x)p^2 - m^2]^2} \\ &= \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 - (x(1-x)p^2 - m^2)]^2} \\ &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 + \Delta]^2} \end{aligned}$$

$$\begin{aligned} &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \int_0^1 dx \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(2-d/2)}{\Gamma(2)} \Delta^{d/2-2} \\ &= i \frac{\hat{g}^2 \mu^{2\epsilon}}{2} \frac{1}{(4\pi)^{3-\epsilon}} \Gamma(\epsilon-1) \int_0^1 dx \Delta^{1-\epsilon} \end{aligned}$$

Now we have to identify the divergent term, which clearly stems from the $\Gamma(\epsilon-1)$, as we have

$$\begin{aligned} \mu^{2\epsilon} &\approx 1 + 2\epsilon \ln \mu + \mathcal{O}(\epsilon^2), \\ (4\pi)^{-\epsilon} &\approx 1 - \epsilon \ln 4\pi + \mathcal{O}(\epsilon^2), \\ \Gamma(\epsilon-1) &= \frac{\Gamma(\epsilon)}{\epsilon-1} \approx -\frac{1}{\epsilon} + \gamma_E + \mathcal{O}(\epsilon), \\ \Delta^{-\epsilon} &\approx 1 - \epsilon \ln \Delta + \mathcal{O}(\epsilon^2). \end{aligned}$$

Therefore, we obtain

$$i\Sigma_2^{\text{div}} = -i \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon} \int_0^1 dx [-x(1-x)p^2 + m^2] = -i \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon} \left[m^2 - \frac{p^2}{6} \right].$$

Using the minimal subtraction renormalization, we finally obtain

$$\begin{aligned} Z_\phi &= 1 - \frac{\hat{g}^2}{12(4\pi)^3} \frac{1}{\epsilon}, \\ Z_m &= 1 + \frac{\hat{g}^2}{2(4\pi)^3} \frac{1}{\epsilon}. \end{aligned}$$

We can use a similar procedure for the renormalization of the 3-point function. Here we denote

$$i\Gamma_3^{(1)}(p_1, p_3) = i(g + \Sigma(p_1, p_3))$$

and compute

$$\begin{aligned} i\Sigma &= (-ig)^3 \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} \frac{i}{(k+p_1)^2 - m^2} \frac{i}{(k-p_3)^2 - m^2} \\ &= g^3 \mu^{3\epsilon} \Gamma(3) \int_0^1 dx \int_0^1 dy \int_0^1 dz \int \frac{d^d k}{(2\pi)^d} \delta(1-x-y-z) \frac{1}{(\dots)^3} \\ &= 2ig^3 \mu^{3\epsilon} \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{[k_E^2 + \Delta]^3}, \end{aligned}$$

where we have defined

$$\Delta = m^2 - [y(1-y)p_1^2 + (1-x-y)(x+y)p_3^2 + 2y(1-x-y)p_1 \cdot p_3].$$

Now, we have

$$\begin{aligned} i\Sigma &= 2ig^3\mu^{3\epsilon}\frac{1}{(4\pi)^{d/2}}\frac{\Gamma(3-d/2)}{\Gamma(3)}\int_0^1 dx \int_0^{1-x} dy \Delta^{d/2-3} \\ &= ig^3\mu^{3\epsilon}\frac{1}{(4\pi)^{3-\epsilon}}\Gamma(\epsilon)\int_0^1 dx \int_0^{1-x} dy \Delta^\epsilon. \end{aligned}$$

Here again using the $\overline{\text{MS}}$, we extract only the divergent term. It is worth noting that due to the logarithmic character of the divergence, the divergent term does not depend on the specific form of $\Delta(p_1, p_3, m)$, and therefore the precise computation of the Feynman parametrization could have been omitted.

$$i\Sigma_3^{\text{div}} = g^3 \frac{1}{(4\pi)^3} \frac{1}{\epsilon} \int_0^1 dx \int_0^{1-x} dy 1 = \frac{g^3}{2(4\pi)^3} \frac{1}{\epsilon},$$

which gives the following renormalization term

$$Z_g = g - \frac{g^3}{2(4\pi)^3} \frac{1}{\epsilon}.$$